

HYPERTROPHIC CARDIOMYOPATHY AND RELATED DISORDERS

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Hypertrophic cardiomyopathy (HCM) is a myocardial disorder characterized by the presence of excessive left ventricular thickness in the absence of conditions causing physiologic hypertrophy such as hypertension, valvular aortic stenosis, or the athletic heart. Hypertrophy is severe and usually asymmetric with the ventricular septum disproportionately involved in more than 90% of cases. Echocardiography usually establishes the diagnosis. Cardiac magnetic resonance (CMR) imaging is often used to confirm the diagnosis or provide additional prognostic or anatomic information. A maximum left ventricular thickness of more than 15 mm anywhere in the ventricle is diagnostic of HCM, or more than 13 mm if a genetic mutation known to cause HCM is present.^{1,2} Viewed microscopically, the myocardium demonstrates a characteristic pattern of myocyte fiber disarray and fibrosis. HCM is a genetic disorder with autosomal dominance and variable penetrance. As the most common form of inherited heart disease, it is estimated to affect between 1 in 200 and 1 in 500 persons.¹ Mutations in at least 11 genes coding for cardiac muscle contractile proteins have been described; the great majority of these mutations involve the beta-myosin heavy chain and the myosin-binding protein C. The clinical presentation and natural history of HCM vary widely, from a relatively benign and asymptomatic condition to one characterized by disabling symptoms, heart failure, or sudden cardiac death (SCD).¹⁻⁴

Interesting hemodynamic consequences arise in patients with HCM, depending on whether or not left ventricular outflow tract obstruction (LVOTO) develops. Hemodynamically, this manifests as a dynamic pressure gradient between the left ventricle (LV) and the aorta. HCM is often classified as *obstructive* or *nonobstructive*, based on the presence of this finding. When present, the dynamic nature of obstruction may cause marked variability in pressure gradients depending on the loading conditions and contractile state of the heart. Gradients may be present at rest with variable degrees of obstruction for as many as one-third of patients with HCM, but the obstruction may also be *latent*, indicating that it is absent at rest and present only after provocation in up to an additional one-third of patients. The nonobstructive forms of HCM lack resting or provokable gradients but may be associated with hemodynamic abnormalities from the profound diastolic abnormalities often present in this condition. Mid-cavitary obstruction is present in around 3% to 5% of patients and is often associated with left ventricular aneurysm formation and a worse prognosis.

PATHOPHYSIOLOGY

Excessive and inappropriate increases in left ventricular thickness lead to several important consequences (Table 7.1). The thickened myocardium impairs left-ventricular compliance, causing elevations in the left-atrial pressure, left-atrial size, and pulmonary artery pressures. Sarcomere mutations alter the configuration of myosin heads during diastole, shifting from the physiologic “super-relaxed” state into the more inefficient “disorder-relaxed” state. This directly reduces cardiac lusitropy and increases basal myocardial energy expenditure. Diastolic dysfunction from all three causes—abnormal compliance, abnormal tissue relaxation speed, and abnormal energy expenditure—can lead to dyspnea and decreased exercise intolerance.

Arrhythmias frequently complicate the course. Atrial fibrillation is the most common comorbidity and ventricular tachycardia is the most feared. Atrial fibrillation may be poorly tolerated because of the associated diastolic dysfunction and the greater importance of the atrial contribution to cardiac output. Importantly arrhythmias (particularly ventricular) represent the primary mechanism of SCD in HCM. An increase in myocardial oxygen consumption from excessive hypertrophy outstrips myocardial blood supply, potentially causing myocardial ischemia and triggering arrhythmia. This may occur despite normal coronary arteries. In addition, ischemia may originate from reduced coronary flow reserve from abnormalities of the small resistance vessels present in the hypertrophied myocardium. Systolic compression of the coronaries (due to intramyocardial bridging) has been observed and may serve as yet another cause of ischemia. In the setting of these ischemic triggers, underlying fibrosis then provides a robust substrate for ventricular tachycardia.

One of the most important physiologic abnormalities in HCM is the presence of LVOTO. A resting or provokable gradient is observed in about 25% to 60% of cases.^{1,5,6} The mechanism of obstruction had previously been a focus of considerable debate. Extensive investigations over the years have established the role of both systolic anterior motion (SAM) of the mitral valve leaflet and of mitral valve abnormalities themselves as the cause of outflow tract obstruction.⁵⁻⁸ Severe hypertrophy distorts the normal relationship between the mitral apparatus and the muscular outflow tract. The asymmetrically hypertrophied proximal intraventricular septum narrows the orifice of the outflow tract. Distortion of the ventricle causes redundancy of the mitral leaflets, leading to SAM of the leaflets, causing obstruction

Table 7.1 Pathophysiologic and Clinical Consequences of Hypertrophic Cardiomyopathy	
PATHOPHYSIOLOGIC ABNORMALITY	CLINICAL SYNDROME
Diastolic dysfunction	Dyspnea
	Exercise intolerance, fatigue
	Arrhythmia
Myocardial ischemia	Angina
	Dyspnea
	Arrhythmia
Obstruction	Dyspnea
	Exercise intolerance, fatigue
	Chest pain
	Syncope
	Endocarditis
	Arrhythmia

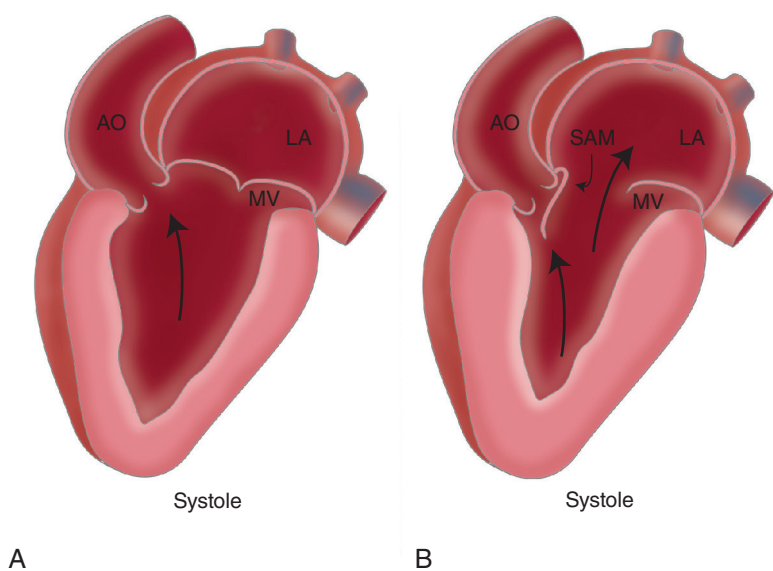


Fig. 7.1 (A) In normal patients the mitral valve closes during systole, leaving the outflow tract unobstructed. (B) In patients with hypertrophic obstructive cardiomyopathy, distortion of ventricular geometry and severe septal hypertrophy cause systolic anterior motion of the mitral valve. This is the mechanism of outflow tract obstruction in hypertrophic obstructive cardiomyopathy and is associated with significant mitral regurgitation. AO, Aorta; LA, left atrium; MV, mitral valve; SAM, systolic anterior motion.

via drag force (Fig. 7.1). It was thought that turbulence within the outflow tract led to a Venturi effect, drawing the anterior mitral leaflet into the outflow tract; however, SAM begins prior to the onset of ejection, suggesting that the Venturi effect contributes little to the outflow tract obstruction. SAM of the mitral valve also results in mitral regurgitation, explaining the frequent finding of significant mitral regurgitation in the presence of outflow tract obstruction. Outflow tract obstruction causes symptoms of dyspnea, fatigue, exercise intolerance, anginal chest pain, and syncope.

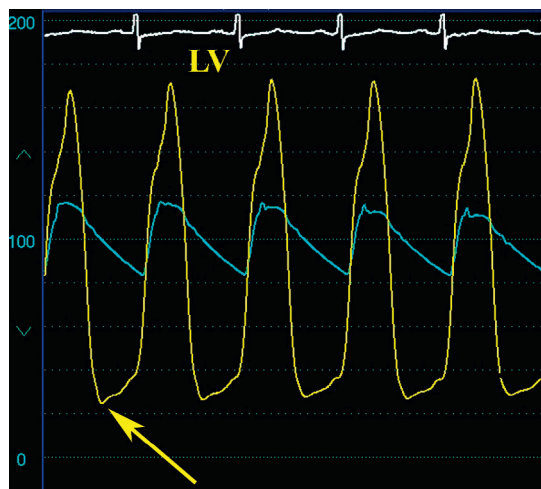


Fig. 7.2 Diastolic dysfunction is common in hypertrophic cardiomyopathy. Abnormal relaxation is manifested by elevation of the early diastolic pressure (arrow), in this case measuring 20 mm Hg. LV, Left ventricle.

HEMODYNAMIC ABNORMALITIES

A wide spectrum of hemodynamic findings may be observed in patients with HCM depending on the consequences of severe hypertrophy on diastolic function and on the presence and extent of LVOTO.

DIASTOLIC DYSFUNCTION

Impaired relaxation of the LV represents the predominant diastolic abnormality associated with the severe hypertrophy of HCM. The associated increase in chamber stiffness from increased myocardial mass and lower chamber volumes also impairs compliance, further worsening diastolic function.⁵

Impaired relaxation limits rapid, passive left-ventricular filling occurring in early diastole. To compensate for reduced rapid filling, atrial systolic filling is exaggerated. This phenomenon, as well as the decreased compliance itself, is clinically observed by a prominent fourth heart sound on physical examination and by the often dramatic clinical deterioration seen when patients with HCM develop atrial fibrillation or loss of atrial synchrony. Contemporary guidelines place a strong emphasis on rhythm control strategies for this reason.

Abnormal diastolic function correlates with the degree of hypertrophy. Patients with severe resting gradients have the highest elevations of left-ventricular end-diastolic pressure compared with those with latent gradients or no obstruction.⁵ In addition, there may be abnormalities in early diastolic pressure. Normally, left-ventricular diastolic pressure is low in early diastole; patients with HCM may have marked elevations in pressure at the beginning of diastole, reflecting abnormal ventricular relaxation (Fig. 7.2). Reduced compliance of the LV is apparent by the commonly present prominent a wave on the left-ventricular waveform (Fig. 7.3). Left-ventricular diastolic abnormalities elevate the pulmonary capillary wedge and pulmonary artery systolic pressures, and in some cases pulmonary pressures may be markedly elevated, ultimately causing structural changes and increased pulmonary vascular resistance with fixed pulmonary hypertension similar to that observed in other chronic heart failure syndromes.

LEFT VENTRICULAR OUTFLOW TRACT OBSTRUCTION

The dynamic nature of the obstruction in hypertrophic obstructive cardiomyopathy (oHCM) creates characteristic hemodynamic abnormalities distinct from those observed with the fixed obstruction caused by valvular aortic stenosis. In valvular aortic stenosis, fixed orifice obstruction impedes ventricular ejection uniformly throughout systole. This leads to the characteristic delayed upstroke in the aortic pressure waveform and the presence of a large systolic gradient beginning early in systole and peaking at maximum left-ventricular pressure (Fig. 7.4A). In the case of oHCM, no obstruction exists at the onset of ventricular ejection. Instead, obstruction progressively develops during systole as the contractile force of the LV builds and the outflow tract narrows. Thus early in systole, left-ventricular ejection is relatively unimpeded, resulting in a brisk initial upstroke of the aortic waveform culminating in a peak systolic pressure (Fig. 7.4B). As systole continues and obstruction reaches a maximum, aortic pressure drops, and the aortic pressure waveform transforms to appear similar to valvular stenosis with a delayed upstroke. This pattern of initial unimpeded ejection followed by progressive obstruction leads to the characteristic “spike-and-dome” configuration of the aortic waveform in oHCM and is also responsible for the *bifid* pulse on physical exam described in this condition (Fig. 7.5).



Fig. 7.3 A prominent *a* wave (arrow) indicates noncompliance of the left ventricle in a patient with severe, nonobstructive hypertrophic cardiomyopathy.



A



B

Fig. 7.4 There are important hemodynamic differences between valvular aortic stenosis and obstruction due to hypertrophic cardiomyopathy. (A) Fixed obstruction due to valvular aortic stenosis causes a marked delay in the upstroke of the aortic pressure wave. (B) In contrast, early ejection is unimpeded in patients with obstruction due to hypertrophic cardiomyopathy. Progressive worsening during the course of systolic ejection creates a normal initial upstroke and then a delayed upstroke later in systole, resulting in a spike-and-dome morphology.

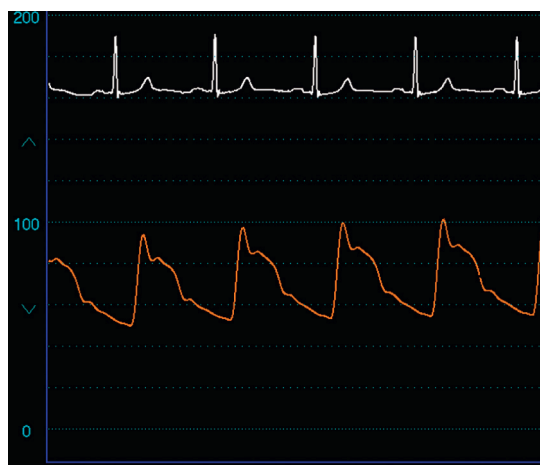


Fig. 7.5 In patients with hypertrophic obstructive cardiomyopathy, the initial upstroke of the aortic pressure wave is normal because early ejection is unimpeded. As obstruction to outflow develops, a fall in pressure occurs, with a spike-and-dome configuration.

Another hallmark of dynamic obstruction, and a clinically useful feature distinguishing oHCM from valvular aortic stenosis, is the drop in the aortic pulse pressure in the first normally conducted beat after the compensatory pause from a premature ventricular contraction (PVC). This finding, known as the *Brockenbrough sign*, was first described over 60 years ago.⁹ This phenomenon is due to the augmentation of the contractile force of the post-PVC beat. In valvular aortic stenosis with *fixed obstruction*, the increased contractile force of the post-PVC beat increases both left-ventricular and aortic pressures. The systolic gradient increases from an associated increase in cardiac output across the valve. Importantly the pulse width on the aortic pressure waveform also increases (Fig. 7.6). In patients with *dynamic obstruction* exemplified by oHCM, the increased contractility of the post-PVC beat worsens the degree of obstruction. This increases the left-ventricular systolic pressure and the systolic gradient but decreases aortic systolic pressure, narrowing the pulse pressure (Fig. 7.7). While it is true that the delayed systole increases preload, which can decrease obstruction in oHCM, the increases in preload, however, do not offset the increased contractility, and thus the observed increase in obstruction post-PVC. Note that the Brockenbrough sign requires only an analysis of the post-PVC aortic waveform. The crucial feature of this sign is simply the narrowing of the aortic pulse pressure on the post-PVC beat. The Brockenbrough sign distinguishes fixed from dynamic obstruction but also helps distinguish true outflow tract obstruction from an intraventricular gradient due to severe hypertrophy and midcavity obliteration or catheter entrapment; the Brockenbrough sign is observed only in the presence of true outflow tract obstruction.

The dynamic nature of the gradient between the LV and the aorta in oHCM reflects the loading conditions of the heart as well as the inotropic state of the myocardium. Significant variations in the gradient are frequently observed at rest.^{10,11} Inspiration causes a decrease in the gradient. It is very common to observe beat-to-beat fluctuations in the gradient without changes in catheter position or respiratory status (Fig. 7.8). It is important to recognize this fluctuation, as a single assessment may underestimate the severity or the true extent of the obstruction potentially present and also may misrepresent the magnitude of gradient reduction after a treatment like alcohol septal ablation.

Patients with latent obstruction have minimal or no gradient at rest. Provocative maneuvers should be considered in those without a resting gradient in whom obstruction is suspected. A variety of techniques have been used; any condition reducing preload or afterload, or increasing contractile force augments dynamic obstruction, unmasking a systolic gradient. Physiologic-based methods are considered the best.¹ In the cardiac catheterization laboratory the simplest method to induce PVCs uses a catheter in the right or left ventricle (Fig. 7.9). Exercise may be used and is more likely to accurately reproduce the hemodynamic condition experienced by the patient's clinical syndrome. This is often difficult to perform in the catheterization laboratory, however, and is best suited for noninvasive assessment by echocardiography.¹ The Valsalva maneuver is helpful in revealing latent obstruction as it increases intrathoracic pressure, lowers preload, and as a consequence augments obstruction.^{1,12} The Valsalva maneuver is an effective method, although it is sometimes difficult for patients under sedation for a cardiac catheterization procedure to comply correctly. To perform the Valsalva maneuver in the cardiac catheterization laboratory, simultaneous left-ventricular and aortic pressures are recorded during the strain and release phases; left-ventricular diastolic and systolic pressures rise during strain, and aortic systolic pressures fall. The provoked gradient appears maximal during and at the end of the strain phase (Fig. 7.10).

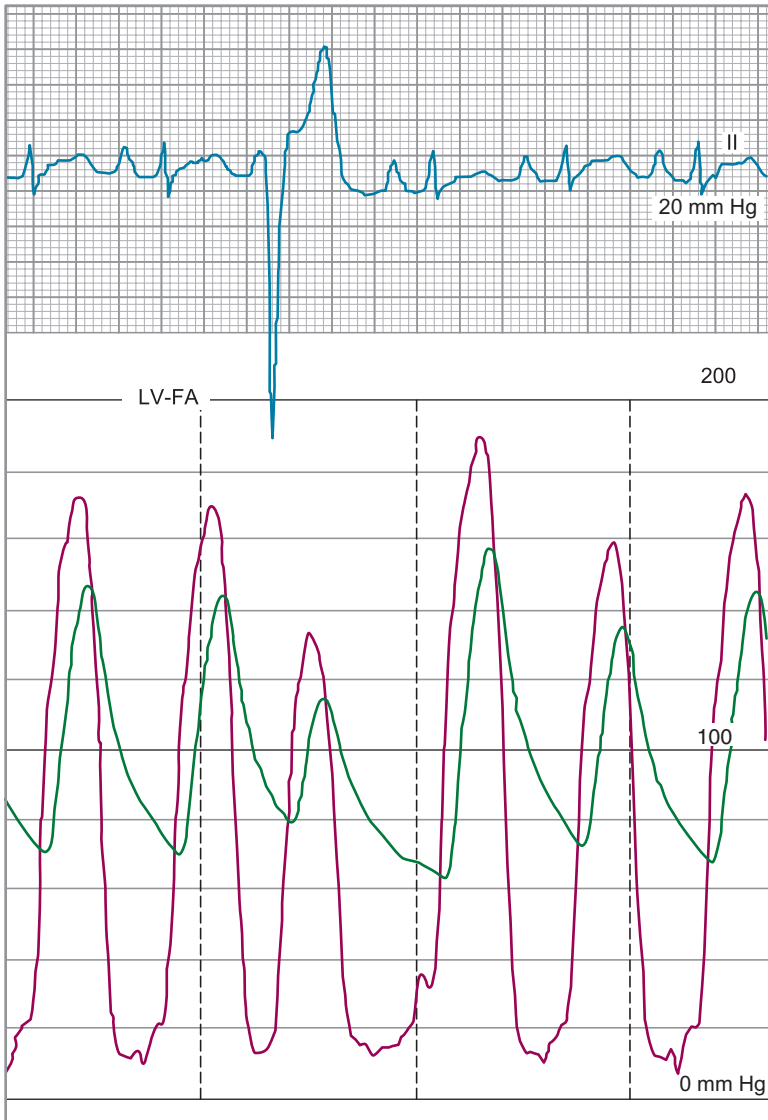


Fig. 7.6 The left-ventricular and aortic pressures (shown here as the femoral artery pressure) both increase in the post-premature ventricular contraction beat in a patient with valvular aortic stenosis, resulting in an increase in the transvalvular gradient and an increase in the aortic pulse pressure. FA, Femoral artery; LV, left ventricle.

Pharmacologic techniques for provocation of a gradient include nitrates to reduce preload and afterload¹² and inotropic stimulation to increase contractility. These techniques are problematic and may not accurately reflect the patient's true physiologic status. In one study a poor correlation existed between the gradient provoked by nitrates and the gradient provoked by exercise; older patients with narrower outflow tracts had a larger gradient provoked by nitrates than exercise, whereas patients who exercised to higher workloads showed higher gradients with exercise than nitrates.¹³ Greater concern relates to the use of direct inotropic stimulation with agents such as dobutamine. It has been well recognized that excessive inotropic stimulation may provoke gradients in normal hearts or in conditions other than HCM such as, for example, in the setting of severe hypertrophy.^{14,15} Thus expert panels do not recommend the use of these agents to decide treatment strategies.^{1,2}

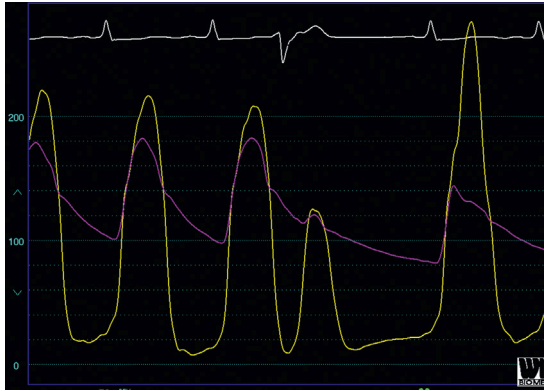


Fig. 7.7 Patients with hypertrophic obstructive cardiomyopathy develop worsening obstruction in the post–premature ventricular contraction (PVC) beat because of augmented contractility. This results in the classic Brockenbrough sign, demonstrated here as a drop in aortic pulse pressure in the post-PVC beat. Also note that the aortic waveform in the post-PVC beat exhibits a spike-and-dome morphology.



Fig. 7.8 Obstruction is dynamic in patients with hypertrophic cardiomyopathy. The left-ventricular-aortic pressure gradient often varies spontaneously, as shown. Note the augmentation of the gradient on the post–premature ventricular contraction beat.



Fig. 7.9 Provocation of the pressure gradient by the introduction of a premature ventricular beat. The resting, peak-to-peak systolic gradient in this patient was approximately 40–50 mm Hg and increased to over 140 mm Hg on the post–premature ventricular contraction beat. Note both the Brockenbrough sign and the classic spike-and-dome configuration on the aortic pressure wave of the post–premature ventricular contraction beat.

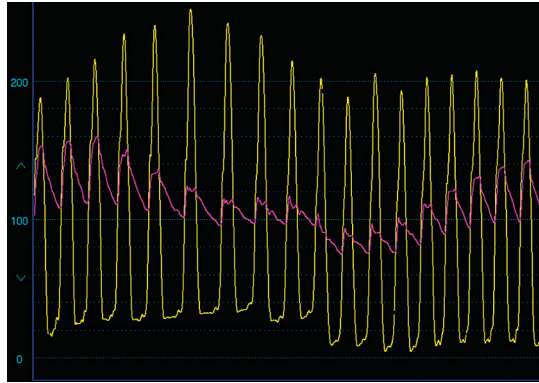


Fig. 7.10 The preload reduction associated with the strain phase of the Valsalva maneuver worsens obstruction and augments the pressure gradient in hypertrophic obstructive cardiomyopathy. On the left side of the panel, the resting gradient is seen and measures about 40 mm Hg. The strain phase of the Valsalva maneuver causes a rise in the left-ventricular diastolic pressure and a marked increase in the gradient.

SPECIFIC CHALLENGES IN HEMODYNAMIC ASSESSMENT

Although most patients with HCM can be accurately assessed noninvasively with Doppler echocardiography, we are often asked to evaluate these patients in the cardiac catheterization laboratory to establish the presence, location, and severity of obstruction. It is important to emphasize that sloppy cardiac catheterization technique, improper catheter position, or inattention to details may confuse the operator and lead to erroneous diagnoses and improper treatment. Specific situations in which invasive assessment is requested include¹⁶ (1) patients with no resting gradient to perform provocative measures (as discussed previously), (2) patients with coexisting mitral regurgitation and concern that the outflow tract Doppler signal is contaminated by a mitral regurgitation signal, (3) distinguishing true outflow tract obstruction from midcavity obstruction, and (4) patients with coexisting aortic valve stenosis to tease out the contribution of serial stenosis.

In the cardiac catheterization laboratory outflow tract obstruction is determined by measuring the pressure gradient between the LV and the ascending aorta. To avoid the potential for catheter entrapment artifact, this is ideally done by positioning an end-hole catheter in the LV inflow tract via a transeptal puncture while simultaneously measuring pressure in the ascending aorta with another catheter.^{16,17} This is not a common practice, due primarily to the increased complexity of performing a transeptal puncture. Thus similar to the catheter configuration used to assess aortic valvular stenosis, most operators determine the pressure gradient with a dual-lumen catheter. In patients with HCM, the optimal catheter to record LV pressure and characterize the outflow tract gradient is one with an end hole and two side holes near its tip (a dual-lumen multipurpose catheter is often used for this purpose). A dual-lumen pigtail catheter is adequate for aortic valvular stenosis but should specifically be avoided when interrogating the left ventricular outflow tract (LVOT) gradient in a patient with HCM. This is due to the fact that a pigtail catheter often underestimates or even fails to identify the gradient in obstructive HCM because several of the catheter's side holes may lie above the obstruction (Fig. 7.11) while the dual-lumen multipurpose catheter with an end hole and two side holes near its tip will more precisely identify an intraventricular pressure gradient, especially in the small ventricular cavities often encountered in this condition.

Assuming the absence of aortic valve stenosis, a gradient between the LV and aorta may not always represent true outflow tract obstruction present at the level of the LV inflow due to SAM of the mitral valve. Another cause of a gradient between the LV and the aorta is midcavity/apical obstruction. If there is severe hypertrophy coupled with hyperdynamic ventricular function and/or reduced ventricular volumes, the LV cavity may be obliterated from the mid-portion of the ventricle to the apex (Fig. 7.12). A catheter tip positioned in that portion of the ventricle will record high ventricular systolic pressures, and thus a systolic gradient will be recorded. In such cases, other features typical of oHCM, such as the Brockenbrough sign or a spike-and-dome configuration on the aortic waveform, will be absent (Fig. 7.13). SAM of the mitral valve will specifically be absent, as well as associated mitral regurgitation. Careful positioning of the catheter in the outflow tract will fail to record a systolic pressure gradient. Performing a transeptal puncture to allow LV pressure measurement precisely at the level of the LV inflow tract may facilitate this assessment.¹⁶

A related pitfall occurs when entrapment of the LV catheter tip occurs within the excessively hypertrophied myocardium. This artifact may provide the appearance of an intraventricular pressure gradient and is often confused with the dynamic obstruction seen in HCM. Several features suggest artifact rather than a true obstruction.¹⁷ The peak LV systolic pressure is seen later in systole, usually at the dicrotic notch of the aortic pressure wave (Fig. 7.14).

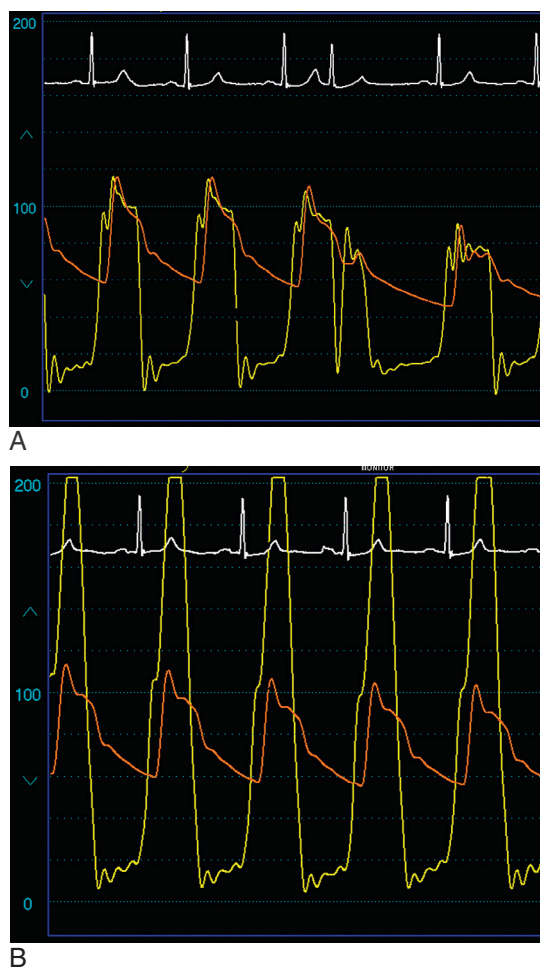


Fig. 7.11 The pigtail catheter is not optimal for the measurement of gradient in hypertrophic obstructive cardiomyopathy. (A) Example of the hemodynamics obtained using a pigtail catheter in the left ventricle in a patient with hypertrophic cardiomyopathy. No gradient is present despite the observation of a spike-and-dome configuration on the aortic waveform and a Brockenbrough sign. (B) The pigtail catheter was exchanged for an end-hole catheter, and a large outflow tract gradient became readily apparent.

Repositioning the catheter tip in the outflow tract at the level of the mitral valve will confirm the absence of a true outflow tract gradient. Often the contour of the ventricular waveform appears bizarre with a narrow, spiked appearance and an initially normal appearance with no gradient early in systole. The Brockenbrough sign and spike-and-dome appearance of the aortic pressure waveform are absent. The LV ejection time is prolonged in true obstruction and decreased with entrapment.¹⁸

Another specific situation where invasive hemodynamic assessment is helpful is in the case of combined valvular aortic stenosis and oHCM. Optimal treatment is dependent on an understanding of the contribution of each component of obstruction. In such cases the operator positions an end-hole catheter in the LV while simultaneously measuring aortic pressure and slowly and carefully withdraws the LV catheter to the aortic valve. This technique can sometimes distinguish the degree of obstruction due to SAM that caused by aortic valve stenosis. Another option is to infuse phenylephrine to increase afterload and thereby reducing SAM. This will reduce or eliminate the gradient from HCM, and any residual gradient can be attributed to aortic valve stenosis.¹⁶ Patients with both valvular aortic stenosis and oHCM may exhibit the classic findings of valvular aortic stenosis with delayed upstroke in the aortic pressure tracing as well as a Brockenbrough sign, indicating dynamic outflow tract obstruction (Fig. 7.15).

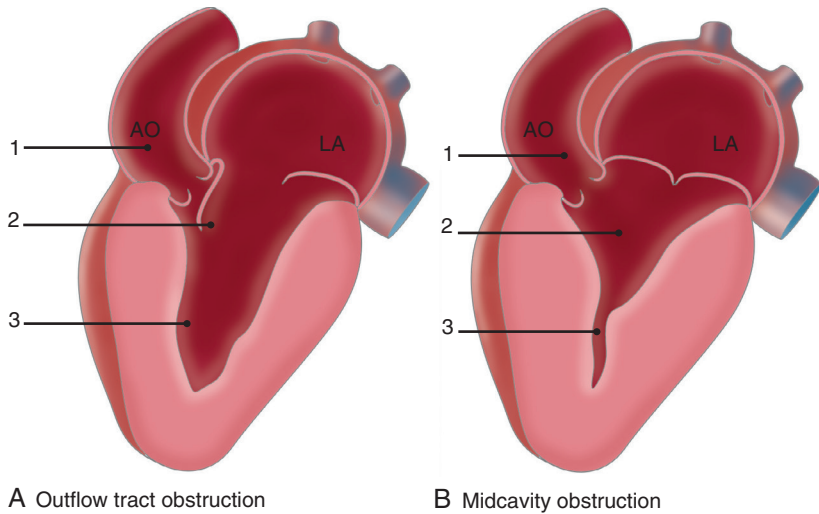


Fig. 7.12 It is important to distinguish true outflow tract obstruction from midcavity obstruction, both of which may demonstrate pressure gradients between the left ventricle and the aorta. (A) True outflow tract obstruction is present due to systolic anterior motion of the mitral valve. Pressure in the ventricular apex (*point 3*) will be higher than the aorta (*point 1*), but a gradient will also exist between the aorta (*point 1*) and the left ventricle at the level of the mitral valve (*point 2*). (B) In contrast, when midcavity obstruction occurs, a gradient will be present between the left-ventricular apex (*point 3*) and the aorta (*point 1*) but not between the aorta (*point 1*) and the left ventricle at the level of the mitral valve (*point 2*). AO, Aorta; LA, left atrium.

The pressure gradients in aortic valve stenosis and from outflow tract obstruction due to HCM can be described in three different forms. The “peak instantaneous gradient” is usually determined by Doppler echocardiography and describes the highest pressure gradient that exists during systole; the “peak-to-peak” gradient is determined by catheterization and is the difference between the peak left-ventricular and peak aortic systolic pressures; and the “mean” gradient is the average gradient present throughout systole. For patients with valvular aortic stenosis the peak instantaneous gradient is higher than the peak-to-peak gradient; for patients with obstruction due to HCM, the peak instantaneous gradient is usually the same as the peak-to-peak gradient (Fig. 7.16). For this reason it is recommended that clinicians report the mean gradient to describe the gradient for aortic valve stenosis for both echocardiography and cardiac catheterization, and that the peak instantaneous gradient is used when describing the obstruction caused by HCM.¹⁹

NONINVASIVE METHODS OF ASSESSING HEMODYNAMICS IN HCM PATIENTS

Although cardiac catheterization served as the basis for hemodynamic evaluation of oHCM for many decades, transthoracic echocardiography (TTE) has become the dominant method for the diagnosis, management, and prognostication in HCM.² In addition to characterizing ventricular structure and function, TTE can assess hemodynamics by using the Doppler methodology to quantify the velocity of blood flow. Pulsed wave (PW) Doppler identifies and measures velocities at a given point in space but is limited by “aliasing,” which does not allow the detection of faster velocities and higher gradients (Fig. 7.17). Continuous wave (CW) Doppler is able to measure higher velocities and gradient, but it does not localize how near or far those structures are to the transducer (Fig. 7.18). Thus using both PW and CW Doppler, and by using the Bernoulli equation to convert blood velocity into pressure gradients, echocardiography reliably quantifies the degree of LVOTG.²⁰ Color Doppler is also helpful to visualize velocities, with red/orange voxels assigned to areas where flow moves toward the probe and blue/violet voxels assigned to areas where flow moves away from the probe (Fig. 7.19).

TTE demonstrates the location, morphology, and degree of LV thickness in patients with HCM. LV thickness may be overestimated when ultrasound beams transect myocardial segments at nonperpendicular angles or when measurements incorrectly include the RV moderator band and may be underestimated when acoustic windows are poor or when traditional views do not capture the thickest segment of the myocardium. The degree of LV thickness directly parallels the risk of SCD, with massive thickness defined as >2.8 to 3.0 cm. Increased LV thickness is the *sine qua non*

of HCM, but the morphologic configuration of the septum directly correlates with the likelihood of identifying a genetic mutation. The so-called “reverse curvature” septum, bowing like a parenthesis into the LV cavity, is associated with a high yield of genetic testing. The isolated basal or sigmoid septum associates with a positive genotype in fewer than 10% of cases likely more reflective of a polygenic predisposition to increased LV thickness rather than a monogenic sarcomere disorder.²¹ Patients with reverse curvature configurations experience increased rates of reduced left ventricular ejection fraction (LVEF) and SCD, whereas the isolated basal configuration is commonly seen in elderly patients and with higher degrees of LV outflow tract obstruction.²² Increased mid-cavitary LV thickness with mid-cavitary obstruction is a particularly high-risk phenotype, with high rates of apical aneurysm formation and sudden death.²³ Apical HCM in isolation tends to have a more favorable prognosis. LVEF, diastolic function, left atrial volume, pulmonary artery systolic pressure, right ventricular thickness, and several other echocardiographic metrics also provide prognostic data.²⁴



Fig. 7.13 This set of waveforms was obtained in a patient with severe hypertrophy to determine if there was outflow tract obstruction. (A) With a dual-lumen end-hole catheter positioned near the left-ventricular apex, no gradient is present between the left ventricle and the aorta, and the aortic waveform has a normal morphology. However, a marked elevation of the left-ventricular diastolic pressure occurs with a prominent a wave. (B) Post-premature ventricular contraction (PVC), there appeared to be provocation of a gradient; however, the aortic pressure wave remained normal in morphology and the pulse pressure was unchanged.

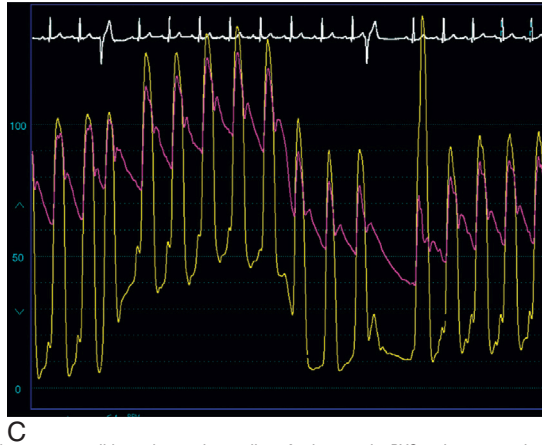


Fig. 7.13, cont'd (C) A Valsalva maneuver did not change the gradient. Again, note the PVC and augmentation of the gradient with the post-PVC beat, but no change in aortic waveform morphology or drop in the aortic pulse pressure occurred. These hemodynamics are most consistent with obstruction in the midcavity location and not true outflow tract obstruction due to systolic anterior motion of the mitral valve.

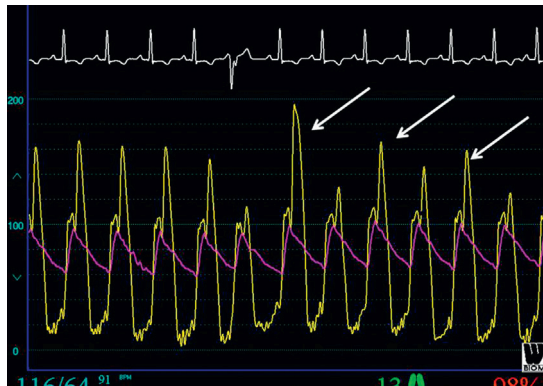


Fig. 7.14 These tracings were obtained using an end-hole catheter in a patient with severe hypertrophy. The left-ventricular pressure tracings have a spiked appearance (arrows) with a peak pressure that occurs late in systole. There is no evidence of the Brockenbrough sign post-premature ventricular contraction. The ventricular pressure tracings likely represent catheter entrapment artifact.



Fig. 7.15 These tracings were obtained in a patient with combined valvular aortic stenosis and hypertrophic obstructive cardiomyopathy. Note the delayed upstroke in the aortic pressure waveform consistent with fixed valvular stenosis. On the post-premature ventricular contraction beat there is a prominent drop in pulse pressure, consistent with the Brockenbrough sign and indicating dynamic obstruction from hypertrophic cardiomyopathy.

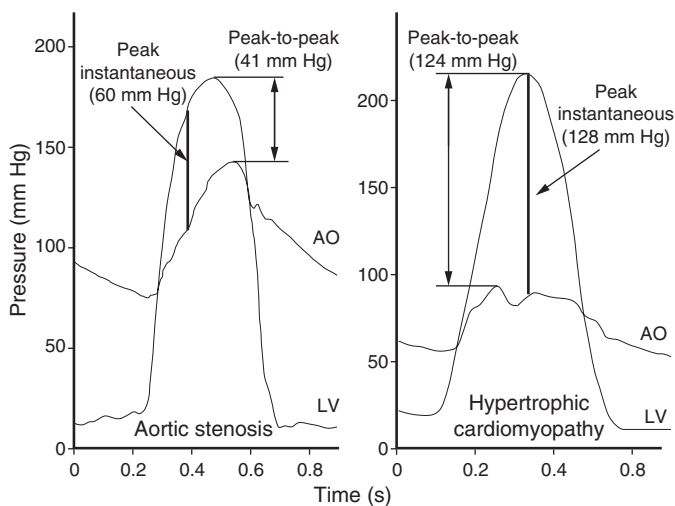


Fig. 7.16 This figure shows the differences between “peak-to-peak” and “peak instantaneous” gradients in aortic valvular stenosis and obstruction from hypertrophic cardiomyopathy. *AO*, Aorta; *LV*, left ventricle. (From Geske JB, Cullen MW, Sorajja P, Ommen SR, Nishimura RA. Assessment of left ventricular outflow gradient: hypertrophic cardiomyopathy versus aortic valvular stenosis. *JACC Cardiol Interv.* 2012;5:675–681.)

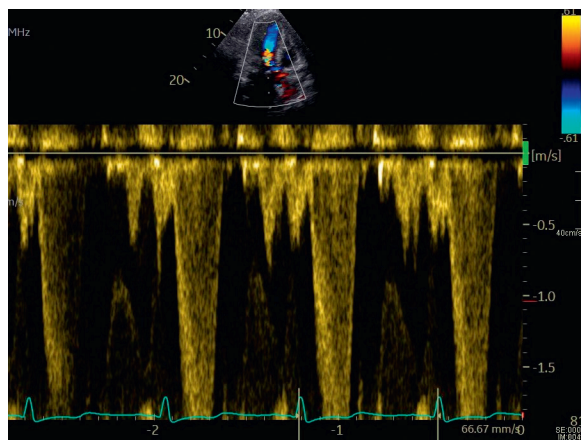


Fig. 7.17 Pulsed wave Doppler in a patient with obstructive hypertrophic cardiomyopathy demonstrating aliasing artifacts at the level of obstruction, thus obscuring the maximum velocity.

Serial imaging with TTE is used during long-term follow-up of patients with HCM, typically at 1- to 2-year intervals. Exercise stress TTE should be considered whenever there are new or worsening symptoms in a patient with a resting LVOTO gradient <50 mm Hg. Compared to TTE, cardiac magnetic resonance imaging (CMR) more reliably assesses LV thickness and LVEF and has superior tissue characterization to help differentiate HCM from phenocopies but tends to have more limited availability. CMR can prognosticate the risk of SCD by assessing the degree of replacement fibrosis or late gadolinium enhancement.²

Assessing LVOTO requires a systematic approach and can be one of the more challenging tasks in TTE image acquisition. First, in two-dimensional (2D) parasternal long, apical three-chamber, or apical five-chamber views, the sonographer should elucidate the degree of SAM of the mitral apparatus. Next, color Doppler imaging should be obtained above and below the level of maximal SAM. When obstruction is present, blood should accelerate and shift from laminar to turbulent flow. This change in the color Doppler pattern with acceleration is akin to the change in

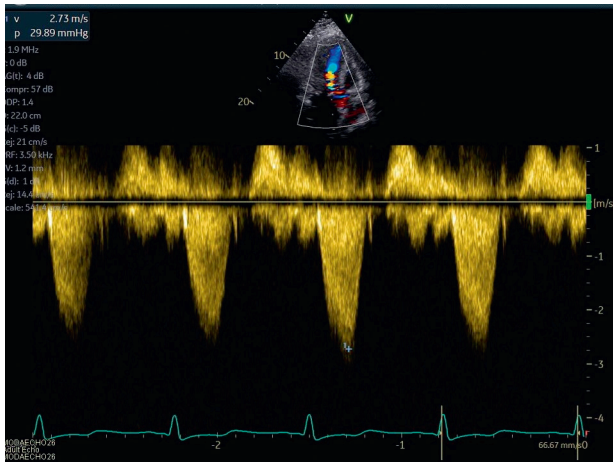


Fig. 7.18 Continuous wave Doppler in a patient with obstructive hypertrophic cardiomyopathy demonstrating a late-peaking systolic profile with a peak velocity of 2.73 m/s, corresponding with a gradient of 30 mm Hg.

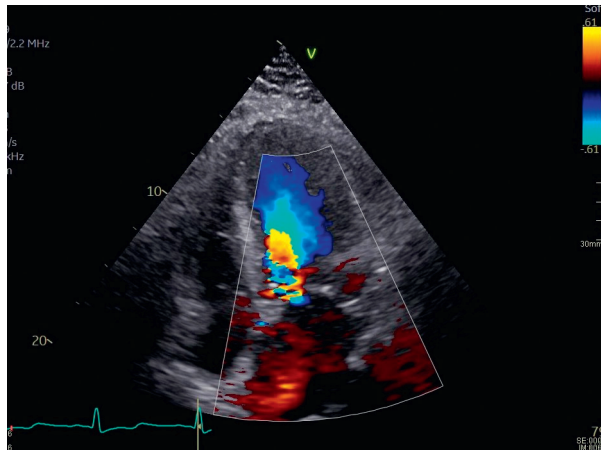


Fig. 7.19 Color Doppler imaging reveals a mosaic pattern of color in the left ventricular outflow tract, consistent with turbulent flow and indicating significant obstruction.

the water leaving the sink. At slow velocities, water falls with a clear, smooth stream. At faster velocities, the stream becomes irregular and rough, with a white-water contour. On color Doppler, turbulent flow results in a mosaic pattern of color. In cases of oHCM, this mosaic pattern should start just prior to the level of obstruction and extend through the LVOT.

At this point in the imaging sequence the narrowest point of the LVOT should have now been identified. PW Doppler imaging should now begin *below* the level of the obstruction in the ventricle. With each successive frame, the PW marker, which indicates from what point in space the velocity is being sampled, should move toward the LVOTO. The reported velocity should alias just proximal to the obstruction, near the area where 2D and color Doppler images demonstrated the greatest narrowing. Subsequently, CW Doppler should sample lateral to the outflow tract obstruction, toward the center of the mitral annulus. With each frame, the sampling line should be shifted toward the LVOTO. The sonographer should be encouraged to take off-axis views to ensure the sampling line is as parallel to the flow through the LVOTO as possible to assess the gradient most accurately. The peak velocity detected with CW will best describe

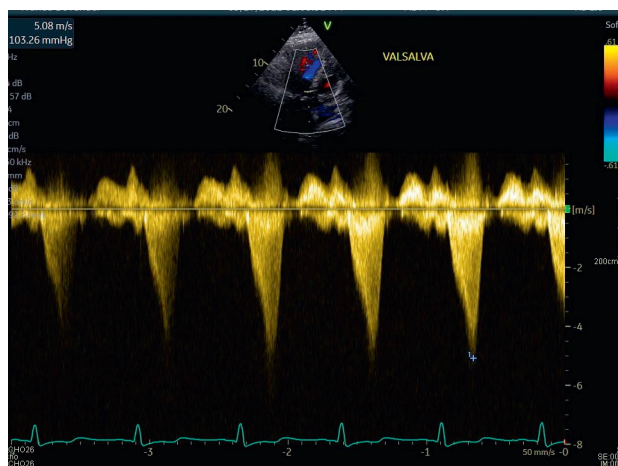


Fig. 7.20 Continuous wave Doppler of the left ventricular outflow tract with Valsalva reveals a peak velocity of 5.08 m/s, consistent with an LVOT gradient of 103 mm Hg. Note the “dagger”-shaped appearance of the time/velocity tracing.

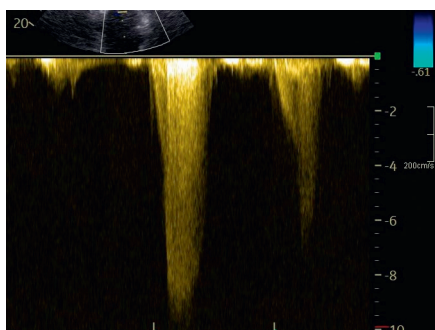


Fig. 7.21 Post-exercise continuous wave Doppler imaging reveals two Doppler profiles. The first, on the left side of the image, has a velocity of 9.3 m/s with a parabolic contour, signifying the mitral regurgitation. The left ventricle (LV) to left atrial (LA) gradient is 346 mm Hg. Assuming an LA pressure of 14 mm Hg, the LV pressure should be about 360 mm Hg. The second profile on the right side of the image is the “dagger”-shaped profile of the left ventricular outflow tract (LVOT). The LVOT profile has a peak velocity of 6.9 m/s, representing an LVOT gradient of 190 mm Hg. With exercise, this patient’s blood pressure was 170/92 mm Hg. The LV to aorta (AO) gradient predicted by the mitral regurgitation Doppler profile is thus 190 mm Hg [(346 + 14) – 170]. The LV to AO gradient predicted by the LVOT envelope alone is also 190 mm Hg. Thus assessments of LV-to-LA gradients via continuous wave Doppler imaging of the mitral regurgitation provide a “check point” on LVOT obstruction assessments.

the LVOTO gradient. Velocity through the LVOT exponentially increases from early to late systole. This exponential rise in velocity is due to the fact that with SAM the radius linearly decreases, which results in exponential decreases in the LVOT area and exponential increases in blood velocity. The time/velocity tracing of the LVOTO should thus have a dagger-shaped appearance (Fig. 7.20).

Last, and perhaps most importantly, the MR jet should be assessed with CW Doppler. The peak velocity of the MR provides the LV-to-LA gradient. By assuming an LA pressure of 15 to 20 mm Hg, the LV-to-LA pressure gradient can provide an estimation of peak LV systolic pressure. LV systolic pressure minus systemic systolic blood pressure should be equivalent to the LVOT gradient. This should provide a double checking of the direct LVOT gradient. The MR Doppler profile classically has a parabolic contour (Fig. 7.21). Given the dynamic, late-peaking nature of SAM-associated MR, MR Doppler profiles can also be dagger shaped similar to LVOT obstruction. The shapes of the Doppler profiles alone cannot reliably distinguish LVOT from MR gradients.

This entire process, from 2D imaging of the SAM through assessment of the MR gradient, should be repeated with Valsalva maneuver. Valsalva maneuvers should be held for at least 5 to 10 seconds prior to imaging, as the initial Valsalva phase increases rather than decreases preload. When there is a mismatch between symptoms and/or visual assessment of the LVOT, the threshold for exercise echocardiography should be low. If the degree of SAM or even the degree of symptoms do not logically match the obtained LVOT gradients even with exercise testing, cardiac catheterization should be considered as the gold standard.

EFFECTS OF TREATMENT

The majority of symptomatic patients with obstruction are treated with medical therapy, traditionally consisting of beta blockers, calcium channel blockers (verapamil or diltiazem), or disopyramide.^{2,25} Although beta blockers improve symptoms, they have shown inconsistent effects on reducing the degree of obstruction or improving diastolic function.²⁶ A recent study found that, in patients with oHCM, exercise was associated with a rise in PCWP that was unaffected by metoprolol, though metoprolol did reduce the gradient, reduce the degree of mitral regurgitation, increase the stroke volume, provide symptom relief, and improve quality of life.^{27,28} Verapamil has been more extensively studied than other calcium channel blockers in this condition and is associated with both modest reductions in the outflow tract gradient and improved diastolic function. Anecdotally, calcium channel blockade may result in fewer side effects, particularly in younger patients. Disopyramide is effective at acutely reducing resting or provoked gradients, with a 50% reduction in gradient sustained over 3 years and no apparent proarrhythmia. In recent years, disopyramide availability has decreased as usage has waned.²⁹

Novel agents are becoming available that may change the natural history of the disease. Mavacamten is an allosteric inhibitor of cardiac myosin-actin interactions, resulting in a reduction in left ventricular contractility with a corresponding reduction in the gradient. It has been shown to reduce the gradient, improve symptoms, improve cardiac biomarkers associated with heart failure, and improve functional status in patients with oHCM. Results in nonobstructive HCM have been less impressive.^{30,31}

Patients with oHCM who suffer continued severe symptoms on maximally tolerated medical therapy (New York Heart Association class 3 or 4) with gradients in excess of 30 to 50 mm Hg at rest or greater than 50 to 60 mm Hg following physiologic provocation are candidates for septal reduction therapy. This can be accomplished either by surgical myectomy or catheter-based alcohol septal ablation.^{2,25} The advantages and disadvantages of these two approaches have been well described and are the source of bountiful and passionate discussion.^{2,32–34} Surgical myectomy is a well-established technique with more than six decades of experience. In the hands of experienced, dedicated surgical centers, surgical myectomy is highly effective at eliminating the outflow tract gradient in more than 98% of patients and is associated with excellent long-term survival, similar to the general population.^{35,36} Although there have been no randomized controlled trials to date comparing the two techniques, numerous registries and meta-analyses have been performed. A few conclusions can be drawn from these studies. Both methods reduce outflow tract gradients and improve symptoms although residual gradients are typically slightly higher with alcohol septal ablation.³⁷ Length of stay and morbidity are higher in patients treated with surgery.³⁸ The 30-day mortality is similar between the two strategies; however, two recent studies derived from large registries found a higher mortality after 4 to 10 years for patients treated with alcohol septal ablation as compared to surgical myectomy even after statistically adjusting for important variables.^{38,39} These were not randomized trials, and the patients treated with alcohol septal ablation were clearly older and had more comorbid conditions; it is possible that the observed difference in mortality was due to unaccounted for variables rather than a true survival advantage for surgery. Not surprisingly, there is a higher rate of heart block requiring permanent pacemaker placement associated with alcohol septal ablation.^{34–36} About 1 in 10 patients undergoing alcohol septal ablation will need a permanent pacemaker as compared to only 1 in 25 patients treated with myectomy.³⁶ Despite achieving similar gradients post procedure, the rate of reintervention is 7.7% for patients undergoing alcohol septal ablation and is only 1.6% for patients undergoing surgical myectomy.³⁶ There remain concerns that alcohol septal ablation might create the substrate for ventricular arrhythmias, potentially leading to late arrhythmic events. While this has not been readily apparent in several studies,^{40–44} it is potentially the explanation for the observation of higher mortality with alcohol septal ablation compared to myectomy in the large, recently reported studies.^{38,39} A randomized controlled trial comparing these strategies is woefully overdue.

The technique for alcohol septal ablation has improved since its inception in the early 1990s. The technique uses conventional angioplasty equipment but requires a sophisticated understanding of septal anatomy and myocardial perfusion. A detailed description of currently used techniques and issues has been described.^{42,45} Appropriate candidates for this technique include patients with “adequate” septal thickness (usually at least 1.5 cm but typically greater than 1.8 cm), class 3 or 4 symptoms despite medical therapy, and at least a 30 mm Hg resting systolic gradient or more than a 50 to 60 mm Hg provokable gradient. The technique depends on the presence of suitable and discrete septal perforators. Ideally the septal perforators should supply only the proximal intraventricular septum at the point of contact between the septum and the mitral leaflet during SAM. This can be confirmed with myocardial contrast echocardiography by selective injection of a myocardial contrast agent into a candidate septal perforator while simultaneously performing a transthoracic echocardiogram (Fig. 7.22). It is important to demonstrate that the septal perforator does not also supply other areas of myocardium, such as the inferior wall, that are not targeted for ablation. If the septal anatomy is appropriate, an appropriately sized, over-the-wire balloon is positioned and inflated to occlude the septal perforator. At this point, with the

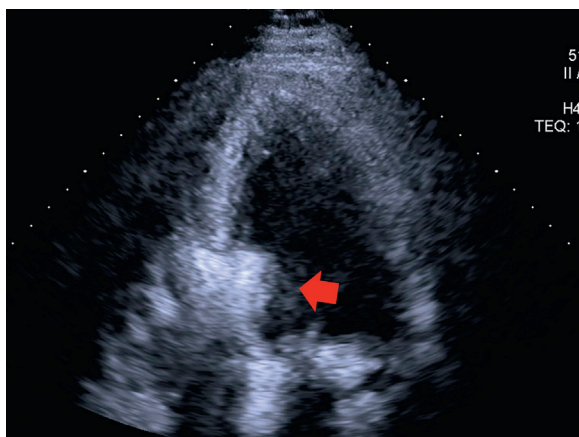


Fig. 7.22 This is an example of myocardial contrast echocardiography used to guide an alcohol septal ablation procedure in a patient with hypertrophic cardiomyopathy and significant obstruction. Contrast was injected into the first septal perforator while echocardiography was performed. Contrast enhancement in the proximal septum is seen (red arrow) at the site of contact with the anterior leaflet of the mitral valve, indicating that the chosen septal perforator will target the correct site of the septum to accomplish a successful alcohol septal ablation.

balloon inflated, a reduction in the gradient may already be observed due to the ischemia caused by balloon occlusion of the septal perforator. The guidewire is removed, and angiographic contrast is injected sequentially first into the lumen of the inflated balloon to ensure no spillage into the left anterior descending artery, and then into the left coronary artery while the balloon is inflated to prove that no antegrade flow occurs around the inflated balloon. The procedure requires the placement of a temporary pacemaker because of the high incidence of transient heart block. Denatured alcohol (1–5 mL) is slowly injected through the lumen of the inflated balloon into the septal perforator and allowed to dwell for 5 to 10 minutes, causing a dense septal infarction and, when successful, elimination of the obstruction. Early in the experience, some operators used a fixed dose of alcohol as high as 3 to 5 mL; however, most experts currently recommend using much lower doses of alcohol to reduce the risk of heart block. The optimal balance between achieving adequate septal ablation and the lowest risk of heart block is a dose of 1.5 to 2.5 mL.^{42,46} The dose should be tailored to the size of the septal artery, and its estimated vascular bed and the thickness of the septum targeted for ablation. The typical hemodynamic effects obtained from alcohol septal ablation are provided in Fig. 7.23. The effects on hemodynamics have been described in a large series of patients who underwent alcohol septal ablation.⁴⁷ An immediate improvement occurs in the outflow tract gradient, which continues to improve over time (Fig. 7.24).

A limitation of alcohol septal ablation is that it may only partially relieve obstruction (Figs. 7.25 and 7.26). One study showed that patients who demonstrate a rapid reduction during balloon occlusion (see Fig. 7.23D) have more gradient reduction with alcohol ablation.⁴⁸

APICAL VARIANT OF HYPERTROPHIC CARDIOMYOPATHY

A rare variant of HCM, first described by the Japanese,⁴⁹ consists of marked hypertrophy that involves only the apex of the LV. Giant negative *T* waves on the precordial leads of the electrocardiogram and the appearance of a characteristic “spade shape” on the left ventriculogram confirm the diagnosis. No specific hemodynamic feature is characteristic of apical HCM. Similar to other patients with severe hypertrophy, diastolic abnormalities may be observed (Fig. 7.27). Because this condition is not associated with LV outflow tract obstruction, no outflow tract gradient, spike-and-dome configuration, or Brockenbrough sign is present. Symptoms such as dyspnea are due to diastolic abnormalities, but the prognosis is usually considered more favorable than other subtypes. In the absence of aneurysm or dense fibrosis the condition is not associated with SCD and carries a relatively benign prognosis, with the major morbidity caused by atrial arrhythmias.⁵⁰

ACQUIRED LEFT VENTRICULAR OUTFLOW TRACT OBSTRUCTION

Dynamic and static LVOTO can be seen in conditions other than oHCM. In cases of dynamic obstruction, the mechanism of obstruction is similar and is due to SAM of the mitral valve. It has been reported in the setting of severe, concentric LV hypertrophy from hypertension or aortic stenosis, particularly under excessive inotropic stimulation or severe volume depletion. In addition, dynamic outflow tract obstruction due to SAM of the mitral valve has been described as a rare occurrence in the setting of an acute anterior myocardial infarction or with myocardial “stunning” in the setting of an acute coronary syndrome when excessive basal left-ventricular hypercontractility is present.^{51–56} In fact, it should be

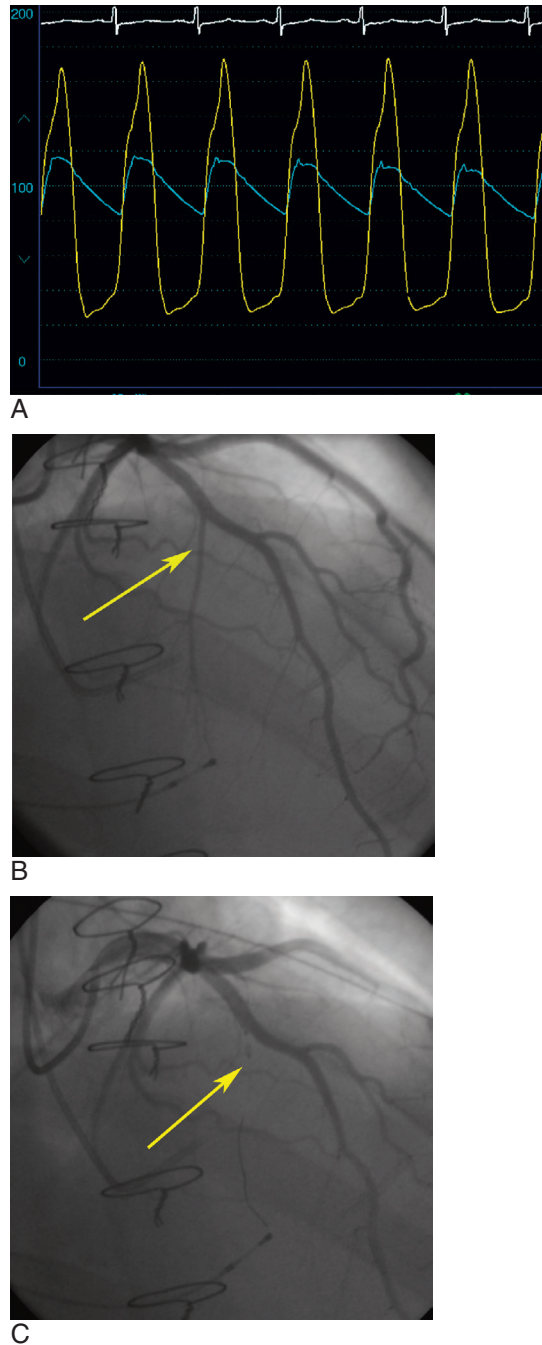


Fig. 7.23 Alcohol septal ablation results in immediate improvement in the hemodynamics of hypertrophic cardiomyopathy. (A) At rest a gradient of nearly 60 mm Hg is present. (B) Coronary angiography confirmed the presence of a suitable first septal perforator (*arrow*). (C) A small 2.0-mm over-the-wire balloon was positioned in this branch and inflated to temporarily obstruct flow (*arrow*).

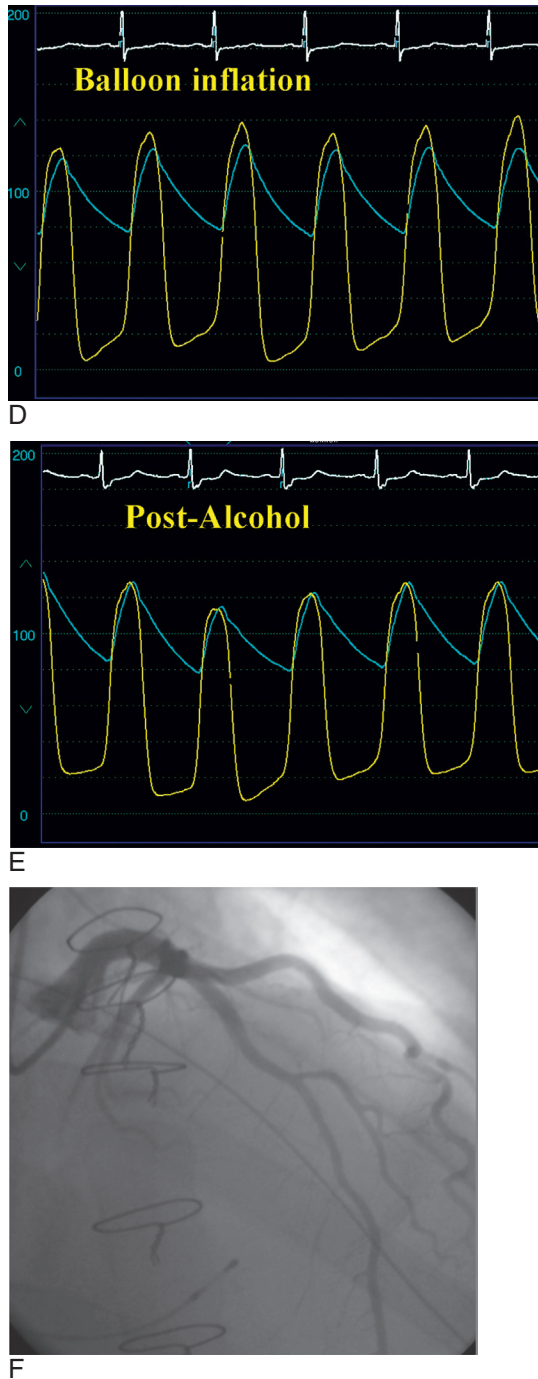


Fig. 7.23, cont'd (D) The pressure gradient during balloon inflation immediately decreased to 10–15 mm Hg. (E) Alcohol was injected through the balloon lumen while the balloon was inflated, and the hemodynamics obtained immediately after alcohol delivery showed resolution of the pressure gradient. (F) The septal perforator typically occludes on the post-procedure angiography in successful cases.

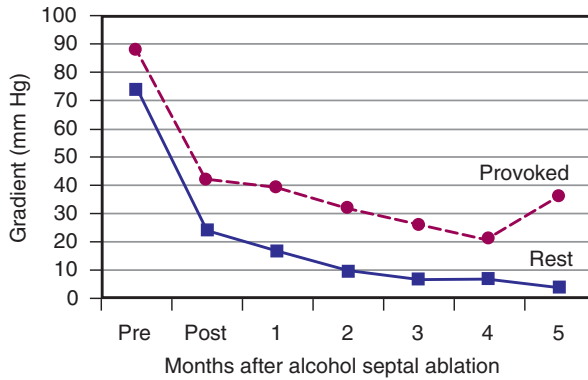


Fig. 7.24 Time course of the pressure gradient changes after alcohol septal ablation. Continued improvement occurs in the pressure gradient after the initial procedure. (From Fernandes VL, Nagueh SF, Franklin J, et al. A prospective follow-up of alcohol septal ablation for symptomatic hypertrophic obstructive cardiomyopathy—the Baylor experience (1996–2002). *Clin Cardiol.* 2005;28:124–130.)

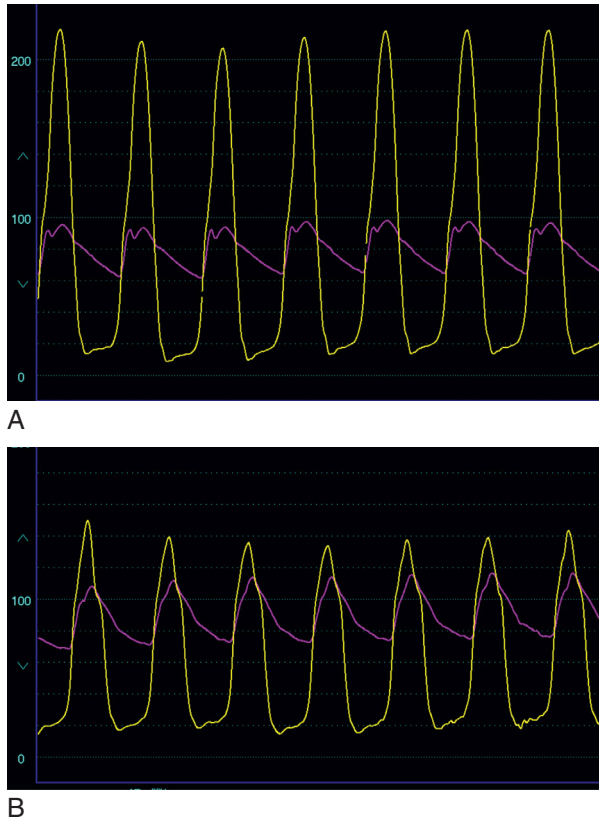


Fig. 7.25 These tracings were obtained in a patient with severe obstruction due to hypertrophic cardiomyopathy who underwent alcohol septal ablation. (A) The resting gradient was more than 100 mm Hg. (B) After alcohol septal ablation the gradient was reduced substantially but was still at about 30 mm Hg at rest, indicating partial relief of obstruction.

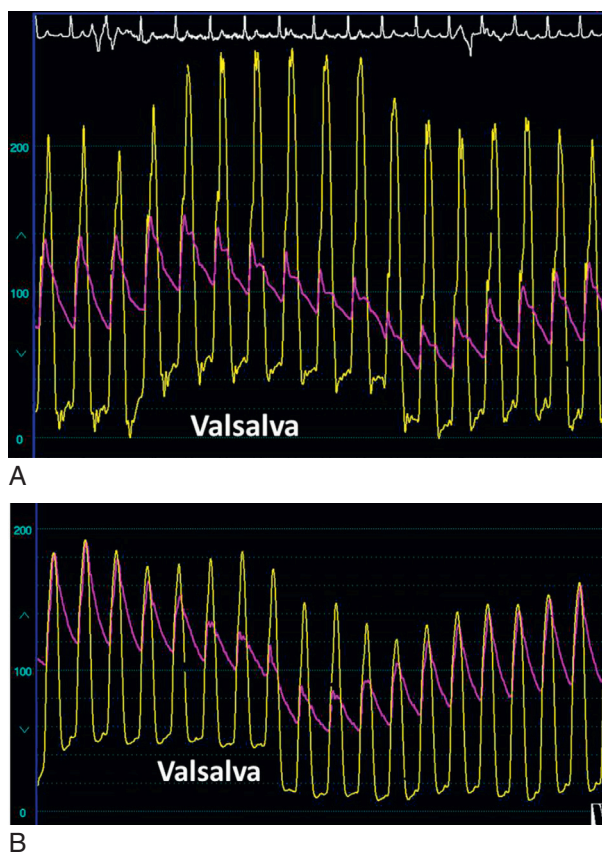


Fig. 7.26 These tracings demonstrate another example of a residual gradient after alcohol septal ablation, apparent only after provocation. (A) Severe obstruction is present at rest, and this is exacerbated by the Valsalva maneuver. (B) Following alcohol septal ablation, simultaneous left-ventricular and aortic pressure tracings show no resting gradient, but a gradient of up to 50 mm Hg is present with the Valsalva maneuver, indicating partial relief of the obstruction.

considered in the differential diagnosis of cardiogenic shock in patients with acute infarction. This phenomenon has been observed in a patient presenting with Takotsubo (or stress-induced) cardiomyopathy presenting with chest pain after grieving over a lost spouse who rapidly developed cardiogenic shock due to outflow tract obstruction (Fig. 7.28).

A very unusual but interesting example of a patient with acquired obstruction is shown in Fig. 7.29. This patient presented to the emergency room with an acute onset of unresponsiveness and profound hypotension, with an electrocardiogram that demonstrated a left-bundle branch block and an emergent, quickly performed bedside echocardiogram that showed hypokinesis of the anterior wall. The patient was thought to be in cardiogenic shock and therefore underwent emergent catheterization, revealing normal coronary arteriography. The hemodynamics suggested outflow tract obstruction. A left ventriculogram confirmed an anterior wall motion abnormality, and repeat echocardiography revealed SAM of the mitral valve as the mechanism of obstruction without evidence of ventricular hypertrophy. The patient was found to have a subarachnoid hemorrhage with neurogenic stunning of the anterior wall and acquired obstruction.

A rare cause of acquired LVOTO is shown in Fig. 7.30. In this case, hypotension and cardiogenic shock occurred early in the postoperative phase of a patient undergoing a mitral valve replacement with a bioprosthetic valve. The valve ring protruded into the outflow tract and caused severe obstruction. Similar occurrences of LVOT obstruction may be observed with transcatheter mitral valve replacement⁵⁷ and more rarely post-transcatheter aortic valve replacement with the so-called “suicide ventricle.”

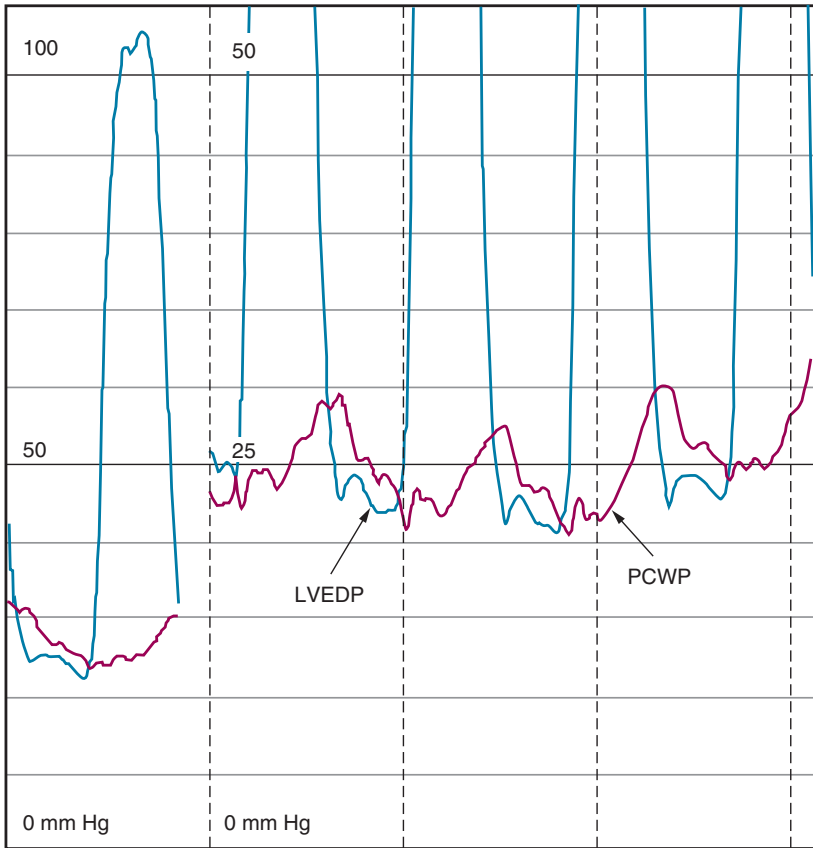


Fig. 7.27 Abnormal diastolic function in a patient with an apical variant of hypertrophic cardiomyopathy. Diastolic pressure is highest early in diastole and represents abnormal relaxation. *LVEDP*, Left-ventricular end-diastolic pressure; *PCWP*, pulmonary capillary wedge pressure.

SUBAORTIC MEMBRANE

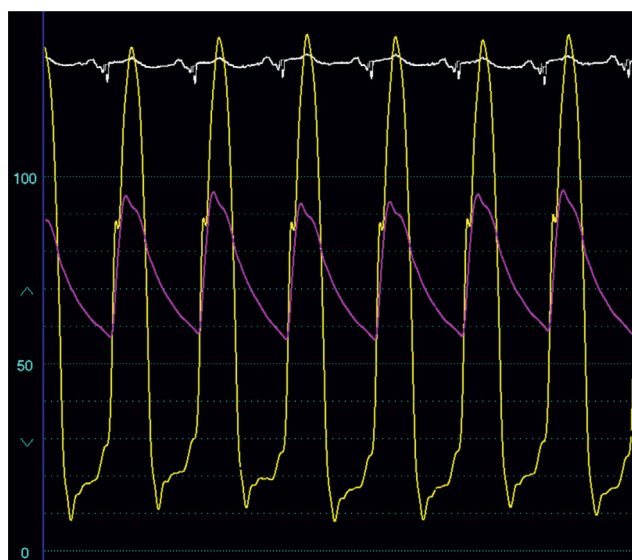
In contrast to oHCM, in which LV obstruction is due to SAM of the mitral valve, the outflow tract obstruction seen in association with a subaortic membrane is caused by the anatomic presence of a discrete membrane, a fibromuscular ridge, or a fibromuscular tunnel that narrows the outflow tract. In addition to the usual symptoms attributed to obstruction, the high-velocity jet may cause progressive and severe aortic regurgitation.

The hemodynamic effects of a subaortic membrane appear similar to those of valvular aortic stenosis. Distinguishing the two conditions requires a careful catheterization technique. The membrane lies only about 1 cm below the aortic valve. If a pigtail catheter is used to assess LV pressure, it will not discriminate the precise location of the obstruction because of the numerous side holes on the catheter. An end-hole catheter is better suited to determine the exact level of obstruction within the outflow tract. Simultaneous aortic and LV pressure waveforms (using an end-hole catheter) document a large systolic pressure gradient. As the ventricular catheter is withdrawn, a pressure gradient within the ventricle will appear to exist just below the aortic valve. No pressure gradient is present across the aortic valve (Fig. 7.31).

Although both oHCM and subaortic membranes exhibit LV outflow tract obstruction, obstruction from a subaortic membrane is *fixed* and thus will behave more like valvular aortic stenosis than oHCM. Similar to valvular stenosis, early ejection will be impaired and no spike-and-dome configuration is present on the aortic waveform, as is seen with dynamic obstruction. The Brockenbrough sign is usually absent, although it may rarely be seen when a predominantly muscular component exists in the membrane.



A



B

Fig. 7.28 (A) The aortic waveform in a patient presenting with cardiogenic shock and chest pain, who was found to have normal coronary arteries and Takotsubo cardiomyopathy. Note the spike-and-dome appearance of the waveform. (B) Simultaneous left-ventricular and aortic pressure tracing confirmed the presence of a gradient in the outflow tract. Echocardiography confirmed systolic anterior motion of the mitral valve caused by geometric distortion of the left ventricle and a normal aortic valve.



Fig. 7.29 Left ventricular outflow tract obstruction may be due to causes other than hypertrophic cardiomyopathy, as shown. This tracing was obtained in a profoundly hypotensive, unresponsive patient taken emergently to the cardiac catheterization laboratory. The coronary arteries were normal. A large gradient is present on simultaneous left-ventricular and aortic pressure tracings, and echocardiography confirmed a large anterior wall motion abnormality and systolic anterior motion of the mitral valve. The patient was found to have a subarachnoid hemorrhage with neurogenic stunning of the left ventricle and acquired outflow tract obstruction. *AO*, Aorta; *LV*, left ventricle.

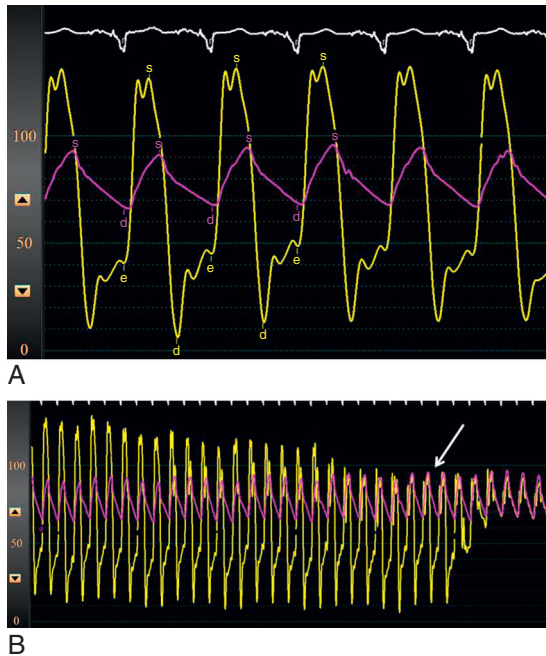


Fig. 7.30 In this case obstruction is caused by a mitral prosthesis protruding into the outflow tract and causing subaortic stenosis. The patient developed cardiogenic shock soon after mitral valve replacement and was brought to the cardiac catheterization laboratory. (A) Simultaneous left-ventricular and aortic pressure waveform show a large gradient and delayed upstroke of the aortic tracing. (B) A slow pullback using an end-hole catheter in the left ventricle demonstrated the obstruction to be at the level of the mitral prosthesis with no gradient across the aortic valve and above the mitral prosthesis (*arrow*). *d*, Diastole; *e*, end diastole; *s*, systole.

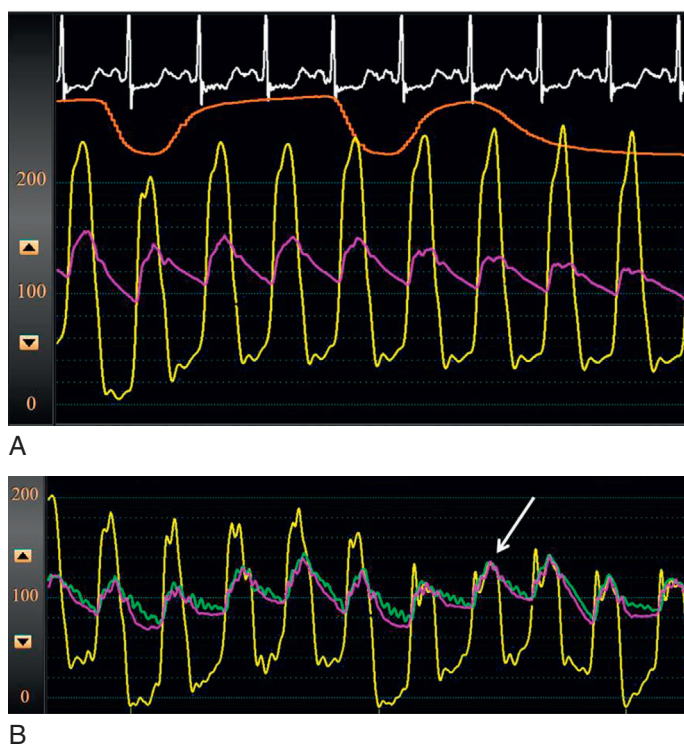


Fig. 7.31 These hemodynamic tracings were obtained in a patient with a subaortic membrane. (A) Simultaneous left-ventricular and aortic pressure waveform shows a large gradient with a delayed upstroke of the aortic tracing. (B) During a slow pullback and using an end-hole catheter in the left ventricle, the subvalvular nature of the gradient can be appreciated by demonstrating no gradient across the aortic valve (arrow).

SUPRAVALVULAR AORTIC STENOSIS

Supravalvular aortic stenosis is a rare condition in which the level of obstruction is above the aortic valve in the ascending aorta. It has classically been described in association with Williams syndrome (elfin facies, peripheral pulmonary artery stenosis, coarctation of the aorta, and supravalvular aortic stenosis) but is actually more commonly observed as an isolated, sporadic form. Rarely it may appear as a familial autosomal dominant form associated with peripheral pulmonary artery stenosis.

The hemodynamic findings of supravalvular aortic stenosis are shown in Fig. 7.32. A gradient is present between the LV and the femoral artery sheath pressure. As the LV catheter is withdrawn across the aortic valve, the absence of a systolic pressure gradient between the LV and the proximal few centimeters of the aorta is apparent. This finding excludes the presence of valvular or subvalvular stenosis. As the catheter is withdrawn further, the precise location of the pressure gradient is identified in the ascending aorta. Coarctation of the aorta exhibits similar hemodynamics, with the systolic pressure gradient localized in the descending aorta at the ligamentum arteriosum.

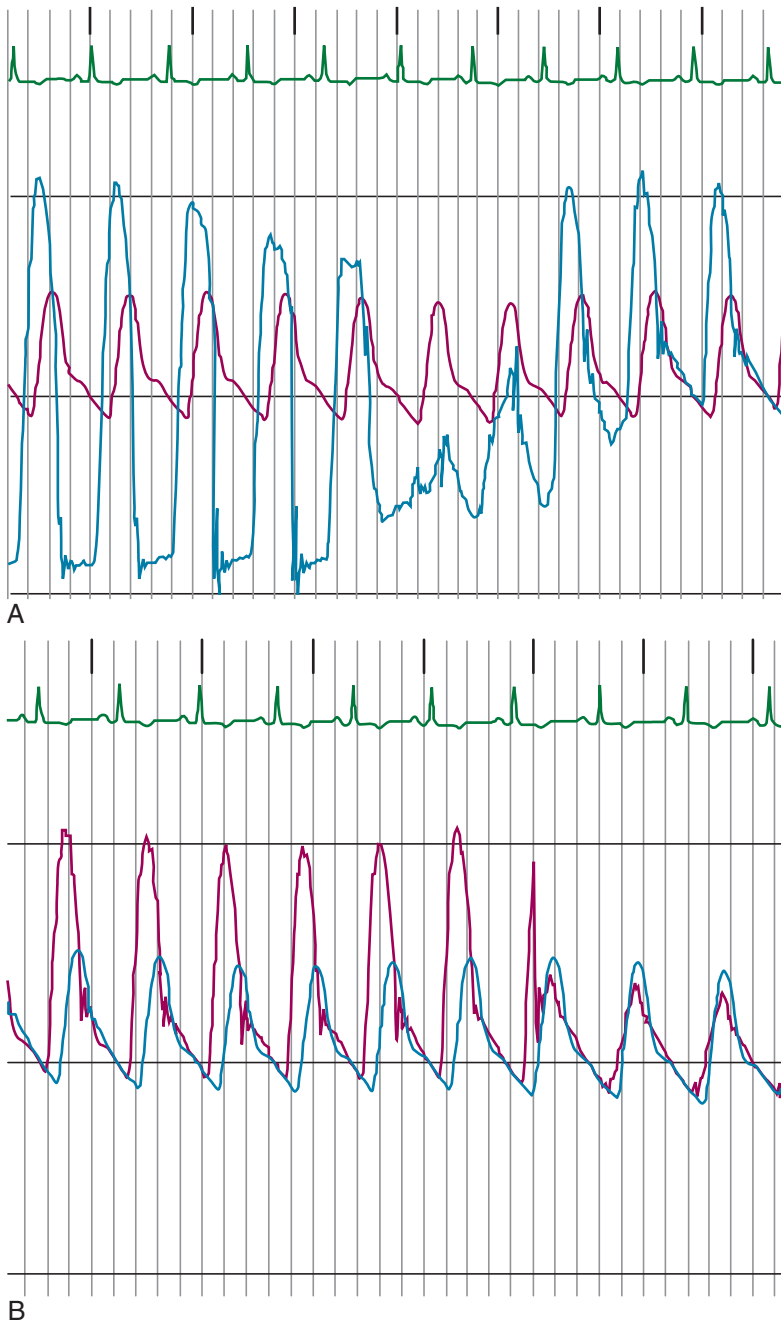


Fig. 7.32 Simultaneous left-ventricular and femoral artery pressure tracings in a patient with supraventricular aortic stenosis. (A) A pressure gradient exists between the left ventricle and the femoral artery (*left of panel*) and the aorta just above the aortic valve and the femoral artery on pullback from the ventricle (*right of panel*). (B) Continued catheter withdrawal above the aortic valve in the ascending aorta discerns the precise location of obstruction.

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